

GOKUL E

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PROFESSIONAL SUMMARY

B.Tech student in Artificial Intelligence & Data Science at Madras Institute of Technology, with hands-on experience building production-grade AI systems and data analytics dashboards. Skilled in computer vision, LLM-based retrieval pipelines, and BI development. Seeking a full-time role to deliver meaningful impact in a data-driven organisation.

EDUCATION

Madras Institute of Technology B.Tech – Artificial Intelligence & Data Science CGPA: 7.5 / 10.0	2022 – 2026
Maharishi Higher Secondary School HSC – 96.6%	2022
Maharishi Higher Secondary School SSLC – 98.2%	2020

SKILLS

Languages: Python, SQL, JavaScript, HTML, CSS
Data & Analytics: Power BI, Excel, DAX, Power Query, NumPy, Pandas
Concepts: DBMS, OOP, Machine Learning, Deep Learning, Computer Vision, RAG, LLMs
ML Frameworks: PyTorch, YOLOv8, OpenCV, LangChain
Database: MySQL, MongoDB

PROJECTS

Continual Learning-Based Smart City Surveillance System

- Tools:** Python, YOLOv8, PyTorch, OpenCV, CNN, YAML, Gradio
- Curated and labelled a custom Indian traffic dataset; fine-tuned YOLOv8 on 75 classes achieving 85.5% mAP@0.5 at 14 FPS real-time inference.
 - Engineered a Continual Learning pipeline with an Experience Replay Buffer to integrate new object classes without catastrophic forgetting.
 - Implemented overlap-based helmet violation detection (89% precision) with automated CSV logging and a zero-downtime model hot-reloading mechanism.

AI-Powered Legal Assistance System

- Tools:** Python, RAG, LLMs, Pinecone, LlamaCloud, LangChain, Streamlit, Pandas
- Built a RAG pipeline over 850+ IndiaCode Acts using Pinecone vector search for semantic, section-level retrieval from unstructured PDF legal documents.
 - Optimized retrieval quality through comparative evaluation of transformer-based embedding models using ranking and relevance metrics, achieving 88% retrieval accuracy with BGE-M3 embeddings.
 - Deployed a LangChain-powered Streamlit interface delivering structured Legal Analysis, Legal Summary, source citations, and auto-generated comparison tables for definitional legal queries.

Library Analytics Portal

- Tools:** Power BI, DAX, Power Query, SQL, Excel
- Consolidated 5 Excel datasets into a Fact Constellation schema (2 fact + 5 dimension tables); implemented DAX-driven KPIs, Top-N ranking measures, and overdue-loan risk heatmaps across 5 library branches.
 - Developed 5-page interactive dashboards enabling management to monitor book circulation, member activity, branch performance, and overdue risks for data-driven decisions..

INTERNSHIP

Industrial AI Intern – Edge Matrix Corporation

June – July 2024

- Fine-tuned YOLOv8 for real-time vehicle detection using transfer learning and data augmentation pipelines.
- Designed and trained CNN models for traffic image classification; integrated Grafana dashboards to monitor inference metrics in production.

CERTIFICATIONS

- [Power BI Data Analyst Professional Certificate – Microsoft \(Coursera\)](#)
- [Excel Skills for Business Specialisation – Macquarie University \(Coursera\)](#)
- [Python 3 Programming Specialisation – University of Michigan \(Coursera\)](#)
- [Machine Learning Specialisation – DeepLearning.AI \(Coursera\)](#)